Randy Tang

rtang0203@gmail.com | 732-666-2746 | U.S. Citizen

SUMMARY

Production backend engineer who owns problems end-to-end — from ambiguous requirements to shipped, reliable systems. 3+ years building high-throughput Python services processing tens of millions of daily transactions, integrating with third-party APIs, and translating stakeholder needs into working solutions. Hands-on AI/ML builder (LLM-powered pipelines, ML data processing, prompt engineering) with a strong foundation in API design and autonomous execution. Cornell CS.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript (building), Java (fundamentals) | **Frameworks/Infra:** Flask, FastAPI, React (building), Node.js (building), Docker, Postgres, SQLite, Git
AI / ML: LLM APIs (Anthropic, Gemini), agentic coding & orchestration, ML data processing & annotation, prompt engineering
Data/Systems: High-throughput pipelines, concurrency, AMPS message queues, Pandas, BeautifulSoup, API design & third-party integrations, WebSocket

EXPERIENCE

Software Engineer — Bank of America Aug 2022 – Current
Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS

- Owned production Python services on the OTC trade reporting engine, processing **tens of millions of transactions daily** across the bank with high reliability and real consequences for failure.
- Designed and shipped an extensible **reference-data framework**: pipeline of services via message queues querying multiple **third-party APIs** per event, with retries, load balancing, and graceful error handling.
- Built a **T+1 trade validation service** checking trade-ID uniqueness across counterparties and issuing corrections at millions of transactions/day — scoped, designed, and shipped independently.
- Developed a parallelized **high-volume backreporting service**, automating a manual process with significant speed and throughput gains — identified the bottleneck and owned the fix end-to-end.
- Operated with **high autonomy** across teams, translating ambiguous requirements from operations and compliance stakeholders into working technical solutions.
- Optimized service reliability and throughput by orders of magnitude; built Python/Pandas tooling to investigate issues across 50–100GB datasets.

Data Analyst (Contract) — Microsoft Mar 2022 – Aug 2022
Viva Topics | Python, Pandas, Jupyter

- Built data processing and annotation pipelines for **production ML models** on the Viva Topics service; cleaned, transformed, and reported on user-collected datasets for the data science team.

Engineering Intern — Northrop Grumman Fall 2019, Summer 2020
NX, ANSYS

- Released 50+ engineering drawings including new designs and large assemblies; performed structural analyses for flight and test components.

PROJECTS

- **Med-Spa Lead-Gen Pipeline** (Python, SQLite, Google Places API, Anthropic API, BeautifulSoup) — End-to-end pipeline discovering businesses via **Google Places API**, enriching web presence, scoring deficiency signals, and generating personalized outreach hooks via **Claude LLM**. 8 idempotent stages, 20 tests.
- **Market Data Dashboard** (Postgres, Docker, Twilio, Grafana) — Ingests live data from **multiple third-party APIs** (Binance, Hyperliquid); automated alerting pipeline with SMS notifications and dashboards.
- **Reading Recommendations App** (Flask, Gemini API, SQLite) — **LLM-powered** daily recommender with prompt engineering, API orchestration, and error handling over a Project Gutenberg corpus.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY Sep 2017 – Dec 2021
B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- **CUAIR (Cornell Unmanned Aircraft)** — Built systems for an autonomous plane with obstacle avoidance, object classification, and airdrop; consistently top team at AUVSI SUAS competition.